

Exact Belief-State Computation at Scale II: The Four-Parameter Cube and the Antichain Census

A computational companion to the belief-state safety-value theory

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Abstract

This paper pushes the exact discipline of belief-state safety values into the curse-of-dimensionality regime: a four-parameter cube with $2^4 = 16$ hidden-parameter cells on 16 stock levels — 256 augmented cells, 17 controls, every computation in exact integer and rational arithmetic. The structure is classified completely. A pairwise-Hamming lemma (every pair of distinct actions sums, over any two parameter cells, to at most twice their agreement count) forces every blind survivable set to consist of Hamming-adjacent cells, and no three cells of the four-cube are pairwise adjacent: no blind policy of any period keeps three or more cells safe. The bands are exact: 16 maximal singletons at the floor's edge, exactly the 32 Hamming-adjacent pairs at every level above it (the matched-singly-mismatched alternation rises at net $\frac{1}{5}$ per two steps with a $\frac{1}{10}$ dip). The observation ladder doubles: each additional exact probe of one parameter doubles the conditional kernel mass — $\frac{1}{16} \rightarrow \frac{1}{8} \rightarrow \frac{1}{4} \rightarrow \frac{1}{2} \rightarrow 1$ at the edge, $\frac{1}{8} \rightarrow \frac{1}{4} \rightarrow \frac{1}{2} \rightarrow 1 \rightarrow 1$ above it — so three probes exhaust observation above the edge. Exact point-based evaluation is verified against definition-level enumeration at seven rational beliefs, three levels, four horizons; the four-step alpha-set deduplicates 83,521 action sequences to at most 545 distinct outcome vectors. The antichain census is the headline: 1,048,576 raw subset evaluations collapse to 496 stored maximal sets. A third instance of the theory's additive deadline law closes the paper (threshold $1 + T/2$, zero probe excursion).

Keywords: exact computation; partially observable control; zero-suppressed sets; survivable sets; Hamming structure; observation design

0.1 Introduction

Two working disciplines govern the exact treatment of belief-state safety values: evaluate exactly where you probe, and store the antichain of maximal survivable sets rather than the lattice (Abaee, 2026b). This paper asks whether the exact discipline survives the curse of dimensionality, and what new structure appears there. The answer is affirmative, and the structure changes: the four-parameter cube is not a larger version of the two-parameter instance but a different regime. Its survivable

sets are governed by Hamming adjacency rather than by coordinate matching; its observation ladder is geometric rather than one-shot; and its raw subset lattice (2^{16} per level) is four orders of magnitude larger than its stored antichain. All claims below are regenerated by the paper's verification script in exact integer and rational arithmetic, standard library only.

0.2 The cube instance

The state is (z, θ) with stock z on the grid $z_0 \in \{1.0, 1.1, \dots, 2.5\}$ and hidden parameter $\theta = (\theta_1, \theta_2, \theta_3, \theta_4) \in \{+1, -1\}^4$: 16 parameter cells on 16 levels, 256 augmented cells. Controls are $u \in \{+1, -1\}^4 \cup \{(0, 0, 0, 0)\}$ (17 actions), and the one-step drift is

$$\dot{z} = d(u, \theta) = -\frac{1}{2} + \frac{1}{5} \sum_{i=1}^4 u_i \theta_i :$$

the matched action $u = \theta$ rises at $+\frac{3}{10}$; a single mismatch gives $-\frac{1}{10}$, two give $-\frac{1}{2}$, three give $-\frac{9}{10}$, the antimatched action $-\frac{13}{10}$, and the hold action $-\frac{1}{2}$ for every cell. The floor is $z \geq 1$. The sum over all sixteen cells of any action's drifts is -8 per step: the cube as a whole always declines, and the blind controller declared here sees no observation of θ .

0.3 The pairwise-Hamming classification

Lemma 1 (pair-sum bound and its survival cost). *Fix the dimension $m \geq 1$, cells $\theta, \theta' \in \{+1, -1\}^m$ at Hamming distance h , the action set $U = \{+1, -1\}^m \cup \{(0, \dots, 0)\}$, the drift $d(u, \theta) = -\frac{1}{2} + \frac{1}{5} \langle u, \theta \rangle$, the floor $z \geq 1$, and the blind class: one action sequence for every parameter branch, of any period and possibly aperiodic. Then: (i) for every $u \in \{+1, -1\}^m$, $\langle u, \theta \rangle + \langle u, \theta' \rangle \leq 2(m - h)$, and the hold action gives 0 for every pair; (ii) if a blind policy keeps both cells above the floor indefinitely, then the pair's long-run average inner product satisfies $\liminf_{T \rightarrow \infty} \frac{1}{T} \sum_{t < T} (\langle u_t, \theta \rangle + \langle u_t, \theta' \rangle) \geq 5$.*

Proof. (i) The coordinates split into $m - h$ agreements and h disagreements; on an agreement $u_i \theta_i = u_i \theta'_i \leq 1$ and on a disagreement the two terms cancel, so the sum is $2 \sum_{\text{agree}} u_i \theta_i \leq 2(m - h)$; the hold action gives 0 termwise. (ii) Survival of cell θ means $z_0 + \sum_{t < T} d(u_t, \theta) \geq 1$ for

every T ; dividing by T and passing to the limit, the Cesàro average of the drifts has $\liminf \geq 0$, and since $d(u_t, \theta) = -\frac{1}{2} + \frac{1}{5}\langle u_t, \theta \rangle$, the $\liminf$ of the average of $\langle u_t, \theta \rangle$ is at least $\frac{5}{2}$; the same holds for θ' , and the two $\liminf$ s add. For a periodic policy the average is the exact cycle average and no $\liminf$ is needed. $\square$

Corollary 2 (the cube's pairs are Hamming-adjacent). *At $m = 4$: (ii) forces the pair average to reach 5 while (i) caps it at $2(4 - h)$, so $5 \leq 2(4 - h)$, i.e. $h \leq \frac{3}{2}$: every blind survivable pair has $h \leq 1$, and since survivability passes to subsets, every pair inside a blind survivable set is Hamming-adjacent. In general the argument gives $h \leq m - \frac{5}{2}$ (Remark 5).*

Lemma 3 (no adjacency triangles). *No three distinct cells of $\{+1, -1\}^m$ (any m) are pairwise at Hamming distance 1. Hence every blind survivable set at $m = 4$ has at most two members, under every blind policy — any period, and no period at all. The bound (i) of Lemma 1 is verified for all 120 cell pairs and all 17 actions at $m = 4$ (and for all 496 pairs and 33 actions at $m = 5$), and Lemma 3 over all $\binom{16}{3} = 560$ triples, by the verification script.*

Proof. Flip coordinates so that the first cell is **1** (flips preserve all Hamming distances). A second cell at distance 1 is **1** with exactly one coordinate i flipped. A third distinct cell at distance 1 from **1** is **1** with exactly one coordinate j flipped: if $j = i$ it is the second cell, and if $j \neq i$ the two differ from each other in the two coordinates i, j , so their Hamming distance is 2. No triangle exists. (Equivalently: the hypercube is bipartite by coordinate parity, so it has no odd cycle.) $\square$

Proposition 4 (bands and maximality). *A singleton $\{\theta\}$ is survivable from every $z_0 \geq 1$ (matched play rises at $+\frac{3}{10}$). A Hamming-adjacent pair $\{\theta, \theta'\}$ is survivable exactly from $z_0 \geq 1.1$: the alternation (θ, θ') gives each member drifts $+\frac{3}{10}, -\frac{1}{10}$ per two steps — net $+\frac{1}{5}$ with a within-cycle dip of $\frac{1}{10}$ — certified at 60 steps, and from $z_0 = 1.0$ the first cell to act under its mismatched action dips below the floor. The maximal survivable sets are therefore exactly the 16 singletons at $z_0 = 1.0$ and exactly the 32 Hamming-adjacent pairs at every $z_0 \geq 1.1$, and nothing larger under any blind policy. An exhaustive search over all 5,219 one-, two-, and three-periodic policies with 60-step certificates reproduces this classification independently.*

Proof. Singletons: matched play has drift $+\frac{3}{10}$ from every $z_0 \geq 1$. Pairs: let $\{\theta, \theta'\}$ be Hamming-adjacent and alternate (θ, θ') ; each member takes the matched drift $+\frac{3}{10}$ and the mismatched drift $-\frac{1}{10}$ once per cycle, a net of $+\frac{1}{5}$ per two steps with the cycle minimum $z_0 - \frac{1}{10}$ after the mismatched step — survivable exactly from $z_0 \geq 1.1$ (the 60-step certificates confirm the cycle arithmetic). No pair from $z_0 = 1.0$: for any first action u , if $u = \theta$ then $\langle u, \theta' \rangle = 2$ and cell θ' falls to $z_0 - \frac{1}{10} < 1$ in one step, while any other u drives at least one member below the floor in its first step (the hold gives $-\frac{1}{2}$ to both); no repair exists because the floor binds at every step. Nothing larger: any set of three or more cells contains two at Hamming distance ≥ 2 (Lemma 3), and that pair violates Corollary 2, so no blind policy keeps it above the floor — and survivability passes to subsets, so the whole set fails with it. Maximality is immediate: singletons extend to pairs from 1.1, pairs extend to nothing. $\square$

Remark 5 (scope in dimension: the ball construction). The pairs-only census is the four-dimensional cube's, and the boundary is sharp. In general dimension the pair argument of Corollary 2 gives only $h \leq m - \frac{5}{2}$; and the constant matched action $u = \mathbf{1}$ gives a cell at Hamming distance k the constant drift $-\frac{1}{2} + \frac{m-2k}{5}$, nonnegative exactly when $k \leq \frac{2m-5}{4}$. Hence the Hamming ball of radius $r^*(m) = \lfloor (2m-5)/4 \rfloor$ around any cell is blind-survivable from every $z_0 \geq 1$: $r^*(4) = 0$, $r^*(5) = r^*(6) = 1$, $r^*(7) = r^*(8) = 2$. At $m = 5$ the radius-one ball — $1+5 = 6$ cells, drifts $+\frac{1}{2}$ and $+\frac{1}{10}$ under constant matched play — survives from the floor's edge, where the four-cube keeps only singletons: the classification changes qualitatively at $m = 5$, and the maximal sets for $m \geq 5$ are not classified here. All eight radii and the sign pattern are verified by the verification script, with 60-step simulations of the radius-one ball at $m = 5$ and the 29-cell radius-two ball at $m = 7$.

Remark 6 (the crude instrument is not sufficient). The averaging identity used on smaller instances — a survivable set S needs $\sum_i |\sum_{\theta \in S} \theta_i| \geq \frac{5}{2} |S|$ — is not sufficient here: the four-set $\{++++, +++-, ++-+, +-++\}$ attains the bound with equality ($4+2+2+2 = 10$) yet is not survivable, since it pairs cells at Hamming distance 2. The pairwise instrument of Lemma 1 is the sharp one.

0.4 The observation ladder doubles

Proposition 7 (geometric ladder). *Prior b_0 is uniform on the sixteen cells; a probe of one parameter refines the support to a subcube. The maximal conditional kernel mass doubles with each probe:*

$$\begin{aligned} \frac{1}{16} &\rightarrow \frac{1}{8} \rightarrow \frac{1}{4} \rightarrow \frac{1}{2} \rightarrow 1 & (z_0 = 1.0), \\ \frac{1}{8} &\rightarrow \frac{1}{4} \rightarrow \frac{1}{2} \rightarrow 1 \rightarrow 1 & (z_0 \geq 1.1), \end{aligned}$$

for zero, one, two, three, and four probes. Above the band's edge three probes exhaust observation — the support is then a Hamming-adjacent pair, jointly survivable, and the fourth probe buys nothing — while at the edge the fourth probe is worth $\frac{1}{2}$, the single-cell mass the pair's dip excludes.

Proof. The companion theory's antichain formula makes the kernel value of a prior the maximal survivable-set mass it charges, and survivability is hereditary downward — a policy keeping a set alive keeps every subset alive — so the value at the uniform prior on a support S is $\max_{A \subseteq S, A \text{ survivable}} |A|/|S|$, attained inside the support. By Proposition 4 the survivable subsets of a subcube are singletons and Hamming-adjacent pairs, so a 2^j -cell subcube has value $2/2^j$ from $z_0 \geq 1.1$ and $1/2^j$ at the edge. The supports after 0, 1, 2, 3, 4 probes of one parameter are subcubes with $j = 4, 3, 2, 1, 0$, giving the two ladders exactly; the fourth edge probe buys the last cell because at the edge only singletons survive and the fully probed support is then itself the surviving cell. The script recomputes all ten values by exhaustive survivable-subset enumeration within each support. $\square$

0.5 Exact point-based evaluation at four parameters

Proposition 8 (exact alpha-set values). *The alpha-vector representation of the underlying theory (its class-restricted*

rational alpha-vectors) evaluated at probed beliefs returns the exact value: for horizons $k = 1, 2, 3, 4$, three stock levels, and seven random rational beliefs, the maximum over the deduplicated alpha-set equals the maximum over definition-level trajectory enumeration — all 84 pairings equal. The four-step alpha-set deduplicates $17^4 = 83,521$ action sequences to at most 545 distinct outcome vectors per level (and to a single vector at the top level’s one-step horizon, where every action is safe — the silent region of the cube). Point-based selection chooses where to evaluate, never what the value is.

Proof. Both sides are maxima of the same linear functional over the same finite set of action sequences: the alpha-set is the image of the sequence set under the map that records, for each parameter cell, whether the sequence survives at the level; the maximum of a function over a finite set equals the maximum over its image, and deduplication removes only repeated vectors, never the attaining one. Exactness of each entry is the definition-level simulation; no discretization enters. $\square$

0.6 The antichain census

Proposition 9 (the antichain, not the lattice). *The raw subset lattice carries $2^{16} = 65,536$ subsets per level — $16 \times 65,536 = 1,048,576$ over the grid — while the stored object is the antichain of maximal survivable sets: 16 sets at the edge and 32 per level above it, 496 stored sets in total. The per-level compression ratio is $4,096\times$ at the edge and $2,048\times$ above it; and by Lemma 3 nothing is lost — the antichain is the whole classification, not a summarization of one.*

Proof. The raw count is $16 \times (2^{16} - 1) = 1,048,560$ nonempty subsets; the headline uses the full lattice $16 \times 2^{16} = 1,048,576$, the larger and therefore conservative figure. The stored count is $16 + 15 \times 32 = 496$ by Proposition 4. The antichain is lossless by the theory’s kernel formula: the value of any prior is the maximal survivable-set mass, read off the stored sets alone. $\square$

0.7 A deadline instance

Proposition 10 (the deadline law on the cube). *Hold for T steps (drift $-\frac{1}{2}$), let the parameter be revealed exactly at T , then play matched forever (drift $+\frac{3}{10}$ on every branch). The companion theory’s additive deadline law applies with blind-drift cost $d_0 = \frac{1}{2}$ and zero probe excursion: viability holds exactly when*

$$z_0 \geq 1 + \frac{T}{2}, \quad T = 0, \dots, 4,$$

verified by direct simulation on the grid. The excursion vanishes because the learned act rises on every branch immediately — the law’s e_p term prices the probe step the free revelation does not need.

Proof. Necessity: under the hold every cell drifts $-\frac{1}{2}$, so at revelation every branch sits at $z_0 - T/2$, and survival to revelation forces $z_0 - T/2 \geq 1$. Sufficiency: the hold keeps every branch at $z_0 - t/2 \geq 1$ for $t \leq T$ exactly under the same inequality, and after revelation the matched action rises at $+\frac{3}{10}$ on every branch, so no later constraint binds. This is the theory’s additive deadline

law at $d_0 = \frac{1}{2}$, $e_p = 0$; the direct branchwise argument needs no scalar-additivity hypothesis, and the simulation certifies the identity cell by cell on the grid. $\square$

0.8 Verification methods

Every number is regenerated by one deterministic script in exact integer and rational arithmetic, standard library only. The battery opens with the pair-sum bound over all 120 pairs and 17 actions; the adjacency-triangle check over all 560 triples; the exhaustive one-, two-, three-periodic classification with 60-step certificates; the ten ladder values by survivable-subset enumeration within each support; the 84 point-based pairings against definition-level enumeration; the census counts; the deadline thresholds by direct simulation; the dimension-scope battery (the pair-sum bound at $m = 5$ over all 496 pairs and 33 actions; the ball radii $r^*(m)$ for $m = 3, \dots, 8$ with the drift sign pattern and 60-step simulations of the radius-one ball at $m = 5$ and the 29-cell radius-two ball at $m = 7$); and every headline number asserted present in this text.

0.9 Delimitations

The blind class is declared and load-bearing: the policy may not depend on the belief, so a single action sequence governs every parameter branch and the pair-sum argument applies; a z -feedback policy could branch its actions across branches (the branches’ stock paths diverge) and is outside the classification, as in the class discipline of the companion theory (Abaee, 2026b). Drifts are deterministic per branch; probes are exact and cost no stock beyond the blind decline they interrupt. The pairs-only classification is scoped to $m = 4$: in dimension five the constant matched action already keeps the six-cell radius-one ball alive from every level (Remark 5), and the maximal sets for $m \geq 5$ are not classified here. Whether some non-blind discipline beats the pairs on the cube is not claimed either way. Scaling beyond sixteen cells — where even the antichain’s input enumeration becomes the bottleneck — is the next step, for which the pairwise-Hamming instrument and the ball construction are the argued generalization targets.

0.10 Conclusion

The exact discipline survives the curse of dimensionality and the cube repays it with structure a pencil would never find: Hamming governance of survivable sets, a doubling observation ladder, a silent region at the top of the stock grid, and a four-order antichain census. The stored object is the antichain; the argued instrument is the pairwise bound; and the two together are what make 1,048,576 subset evaluations collapse to 496 sets without loss.

References

- Abaee, A. (2026a). An obstruction calculus for viability under incomplete observation. Submitted.
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Declarations

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Competing interests

None.

Data availability

No external data were used.

Code availability

The verification script `paper2_exact_belief_computation_v5_verification.py` is publicly available at <https://github.com/MIKEAA2020/general-sustainability> (folder `arena agent 1/paper rewrites/latex`); it reproduces all bounds, bands, ladder values, pairings, census counts, and thresholds verbatim.

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